$$\mathrm{fix} : \underbrace{\mathrm{Tm}\,(\Gamma \triangleright A)\,A}_{\cong \mathrm{Tm}\,\Gamma\,(A \Rightarrow A)} \longrightarrow \mathrm{Tm}\,\Gamma\,A$$

$f\,x : (a \to a) \to a$
$\mathrm{fix}\,f = f(\mathrm{fix}\,f)$

$$\mathrm{fix}\;t = t[\langle \mathrm{fix}\;t\rangle]$$

$t[\langle t[\langle(\mathrm{fix}\,f)\rangle]\rangle] =$
$t[t[t[t[t[\ldots$

$A := \mathrm{Nat} \times \mathrm{Nat}$
input visszafejtés

~~fac = fix(if (zero(fst v0)) then (m0, m1) else (fst v0 − m1, fst v0 * snd v0))~~

kaind: shell, qp rendszer, server, Turing gép, gép, automata, végtelen lista = folyam = stream

---

ind konstr → destr "terminál" ite "véges" Stream : Ty → Ty

kaind destr → konstr "produkál" gen "végtelen"

head : Tm Γ (Stream A) → Tm Γ A

tail : Tm Γ (Stream A) → Tm Γ (Stream A)

genStream : Tm (Γ▷C) A → Tm (Γ▷C) C → Tm Γ C → Tm Γ (Stream A)

head (genStream h t c) = h[⟨c⟩]

tail (genStream h t c) = genStream h t (t[⟨c⟩])

[1, 2, 3, ...] := genStream {Nat} (v0) (suc v0) (m 1)

[3, 3, ...]

head (tail (tail (genStream v0 (suc v0) (m 1)))) = head (tail (genStream v0 (suc v0) (m 2))) =

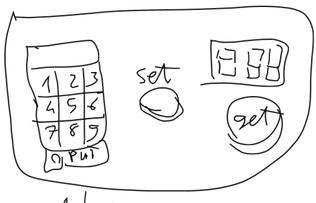
head (genStream v0 (suc v0) (m 3)) = m 3

Machine : Ty
!!
M

put : Tm Γ M → Tm Γ Nat → Tm Γ M

set : Tm Γ M → Tm Γ M

get : Tm Γ M → Tm Γ Nat



genM Tm (Γ▷C▷Nat) C →
Tm (Γ▷C) C →
Tm (Γ▷C) Nat →
Tm Γ C → Tm Γ M

put (genM p s g c) n = genM p s g (p[⟨c⟩][⟨n⟩])

set (genM p s g c) = genM p s g (s[⟨c⟩])

get (genM p s g c) = g[⟨c⟩]

önmagó:
genM {Nat}
(v0 + v1)
(m 0)
v0
(m 0)

VPRI, STEPS project, Alan Kay

compile_a_program;

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t ∘ t ∘ t ∘ t ∘ ---.

t : (N → N) → N → N

t = λn if n=0 then 1 else n * t (n−1)

t f = λn if n=0 then 1 else n * f (n−1)

t (t (t (t (t (... ⊥)

λn. if n=0 then 1 else n * (if n−1=0 then 1 else (n−1) * (if n−2=0 then 1 else (n−2) * ...

nlab coinductive types

W-types

M-types